

Revision — 1.5.2 Moral and Ethical Issues

OCR H446 — one-page revision summary

Must know

- Every issue is a **trade-off**: the same fact benefits one stakeholder and harms another. Always argue **both sides**.
- **Automation & workforce**: raises productivity and creates new jobs, but **displaces** low-skilled workers and can deskill roles. Links to **layoffs** and **offshoring**.
- **Automated decision making** (loans, CV sifting, self-driving): fast, consistent, tireless, handles huge data — but lacks empathy/judgement, is hard to appeal, and raises **liability** questions (trolley problem).
- **AI ethics / bias**: powerful at pattern-finding, but learns and **amplifies bias** in its training data, is a "black box", and accountability is unclear.
- **Environmental**: computing cuts travel (remote work), enables smart grids and paperless working — but causes **e-waste**, huge data-centre and crypto energy use, and rare-earth mining.
- **Censorship & the internet**: protects children and blocks illegal/extremist content — but restricts **free speech** and can be abused by states to silence dissent.
- **Monopolies & open source**: dominance gives standards, polish and support — but stifles competition, raises prices and causes **vendor lock-in**; open source counters this.
- **Digital divide**: the gap between those with/without tech access and skills — online services empower the connected but **exclude** the poor, rural, elderly and disabled, widening inequality.
- **Privacy & surveillance**: mass data collection enables personalised services and crime prevention — but erodes privacy, enables profiling and misuse (links **DPA** and **RIPA**).

Key terms

Term	Definition
Stakeholder	A party affected by a decision (individual, business, government, environment)
Automation	Machines performing tasks once done by humans
Algorithmic bias	Unfair outcomes an AI reproduces/amplifies from biased training data
Digital divide	Gap between those with and without access to technology and skills
Offshoring	Moving work abroad to cut costs
Vendor lock-in	Dependence on one supplier, making switching costly/difficult
Censorship	Controlling what content may be accessed or published
Surveillance	Monitoring/collecting data on people's activity or communications

Worked example / how to — the FOR / AGAINST essay skeleton

For a 9–12 mark **discuss / evaluate** question:

- **Structure:** FOR → AGAINST → FOR → AGAINST → **justified conclusion**. For each point use *point → consequence → stakeholder → counter-point*.
- **Name 3–4+ stakeholders:** individuals/consumers · employees · businesses/employers · government/society · the environment · developing countries/future generations.
- **Develop, don't list:** "It causes job losses" → *why, who is affected, what follows, counter-argument*.
- **Conclusion must take a side and justify it**, often with a condition — e.g. "automate, *but only if* staff are retrained and some staffed tills remain"; "surveillance only if targeted, proportionate and independently authorised".
- **Level 3 (top)** = balanced, multiple stakeholders, both benefits and drawbacks, accurate terms, sustained reasoning and a justified conclusion.

Exam tips

- **Never one-sided** — "discuss/evaluate" needs both sides plus a justified conclusion to reach Level 3.
- **Always conclude** with a clear, reasoned position; don't just stop after listing points.
- Name **specific stakeholders** and **develop** each point rather than listing bald statements.
- Where relevant, **link to the law** (e.g. surveillance → DPA/RIPA) to strengthen reasoning.